

segmentation

Chapter 5

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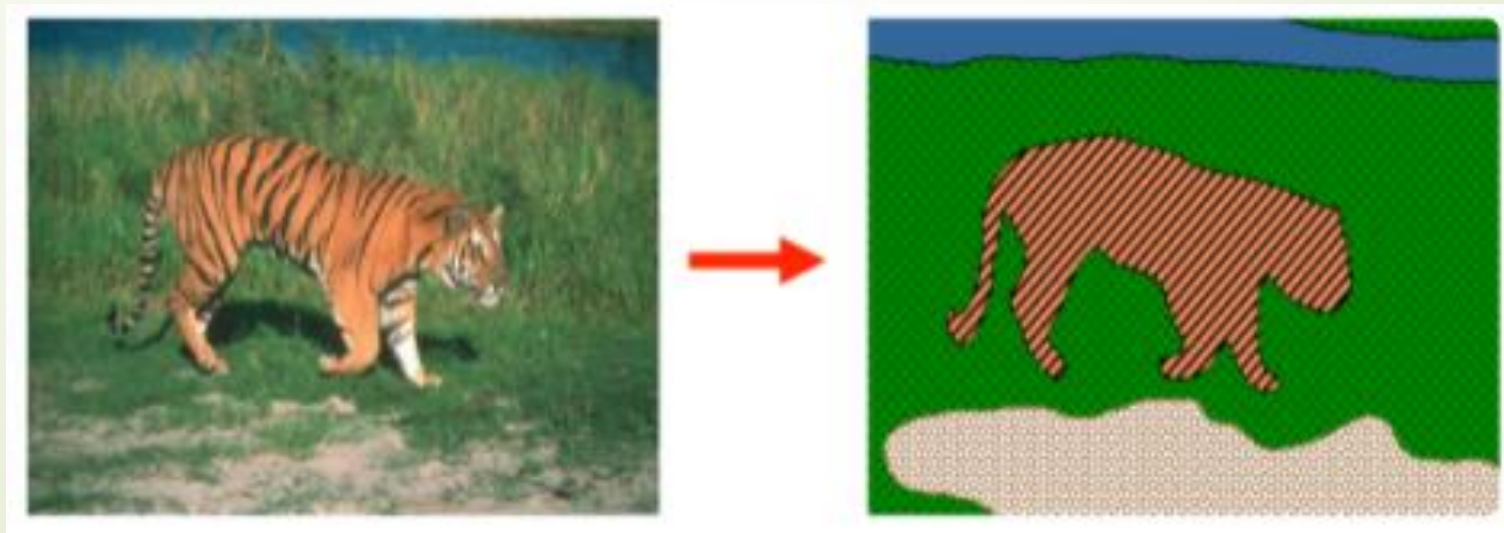
Contents

- ★ Active contours
- ★ Split and merge
- ★ Mean shift and mode finding

Introduction

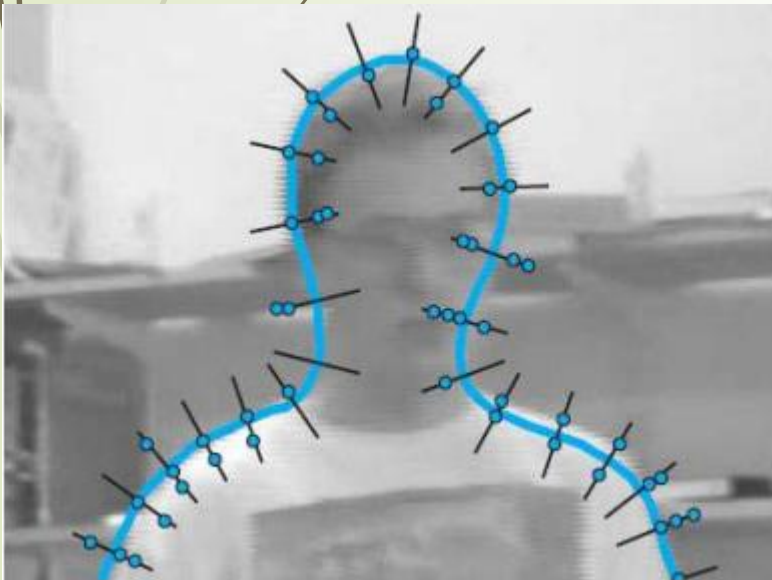
What is image segmentation?

- ★ It involves partitioning a digital image into multiple segments (regions or objects) to simplify and analyze an image by separating it into meaningful components.
- ★ A typical image segmentation task goes through the following steps:
 - Groups pixels in an image based on shared characteristics like colour, intensity, or texture.
 - Assigns a label to each pixel, indicating its belonging to a specific segment or object.
 - The resulting output is a segmented image, often visualized as a mask or overlay highlighting the different segments.



Introduction

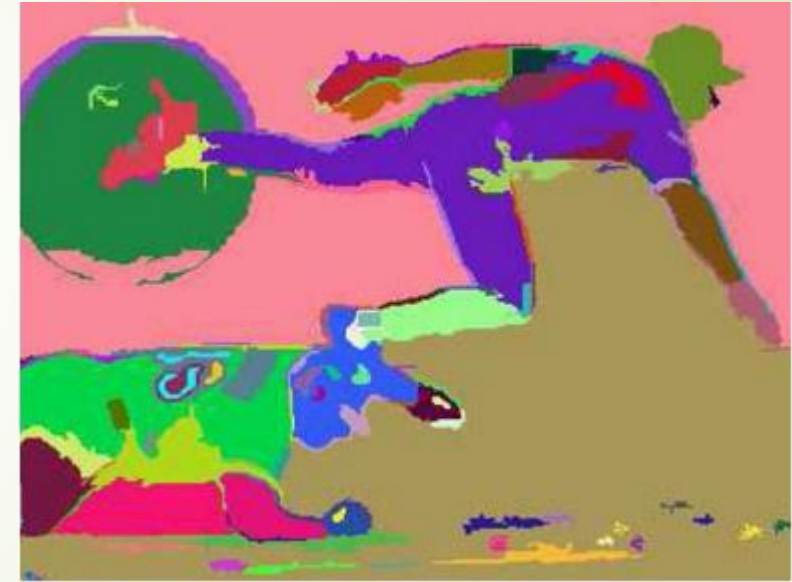
- ★ Various image segmentation techniques exist, such as
 - Active contours, split and merge, watershed,
 - region splitting, region merging, graph-based segmentation,
 - mean shift and model finding, and Normalized cut.



(a) Active Contour



(b) Level sets



(c) Graph based Merging

Introduction



(d) Mean shift



(e) Texture and
intervening contour-
based normalized cuts



(f) Binary MRF solved
using graph cuts

Active Contour

What are active contours?

- ★ Active contour is a segmentation method that uses energy forces and constraints to separate the pixels of interest from a picture for further processing and analysis.
- ★ Active contours are deformable curves that adapt their shape based on the image's features and a set of predefined criteria.
- ★ They can accurately trace object boundaries, even in the presence of noise and irregular shapes.

Active Contour

Why is active contour needed?

- ★ Active contours are mainly used to identify uneven shapes in images.
- ★ They are also used to define smooth shapes in images and construct closed contours for regions.
- ★ Active contour models are employed in various medical applications, particularly for separating desired regions from medical images.

Active Contour

How does active contour work?

- ★ Active contours are the technique of obtaining deformable models or structures in an image with constraints and forces for segmentation.
- ★ Contour models define the object borders or other picture features to generate a parametric curve or contour.
- ★ The curvature of the models is determined using several contour techniques that employ external and internal forces.
 - The *energy function* is always related to the image's curve.
 - *External energy* is described as the sum of forces caused by the picture that is specifically used to control the location of the contour onto the image.
 - *Internal energy* is used to govern deformable changes.

Active Contour (Snake Model)

Snake Model

- ★ The snake model is a technique that can solve a broad range of segmentation problems.
- ★ The model's primary function is identifying and outlining the target object for segmentation.
- ★ It requires prior knowledge of the target object's shape, especially for complicated things.
- ★ Active snake models, often known as *snakes*, are generally configured by using a spline focused on *minimizing energy*, followed by *various forces governing* the image.

Active Contour (Snake Model)

Snake Model

$$\begin{aligned} E_{\text{snake}}^* &= \int_0^1 E_{\text{snake}}(v(s)) ds \\ &= \int_0^1 E_{\text{int}}(v(s)) + E_{\text{image}}(v(s)) + E_{\text{con}}(v(s)) ds \end{aligned}$$

Where, E_{int} represent the internal energy of the spline due to bending

E_{image} gives rise to the image forces

E_{con} gives rise to the external constraint forces

★ The **internal spline energy** can be written

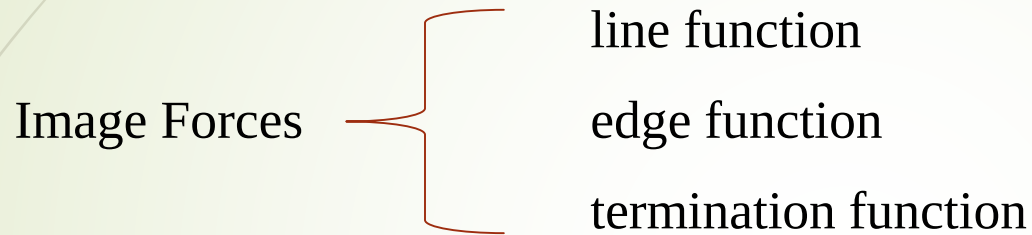
$$E_{\text{int}} = (\alpha(s)|v_s(s)|^2 + \beta(s)|v_{ss}(s)|^2)/2$$

Where, $\alpha(s)$ and $\beta(s)$ are user-defined weights; are first and second-order continuity weighting functions.

Active Contour (Snake Model)

Image Energy

- ★ The image-based potentials are the sum of several terms



$$E_{\text{image}} = W_{\text{line}} E_{\text{line}} + W_{\text{edge}} E_{\text{edge}} + W_{\text{term}} E_{\text{term}}$$

The **line** term attracts the snake to dark ridges

The **edge** term attracts it to strong gradients(edges)

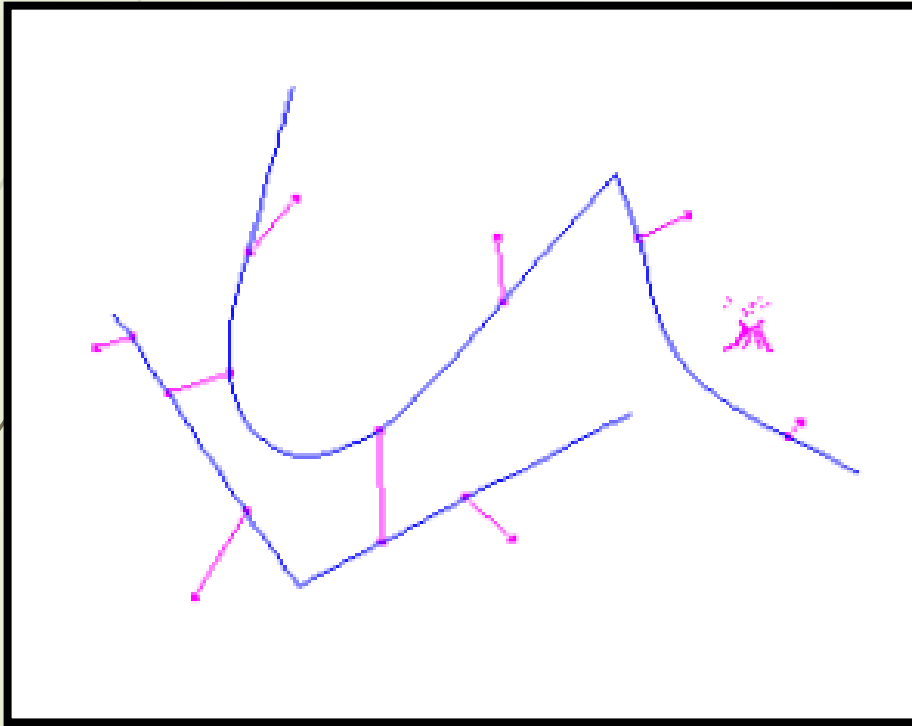
The **term** term attracts it to line terminations.

Constraint Energy

- ★ Allowed for user interaction to guide the snakes, not only in initial placement but also in their energy terms.
- ★ E_{con} can be used to interactively guide the snakes towards or away from particular features

Active Contour (Snake Model)

★ Example



(a) the “snake pit” for interactively controlling shape

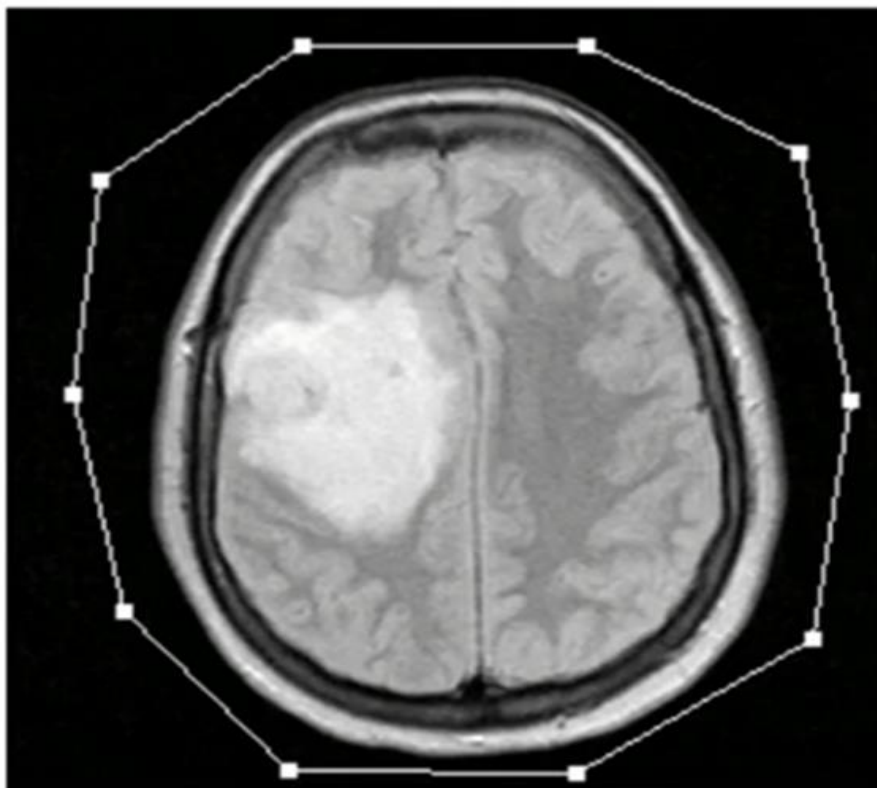


(b) lip tracking

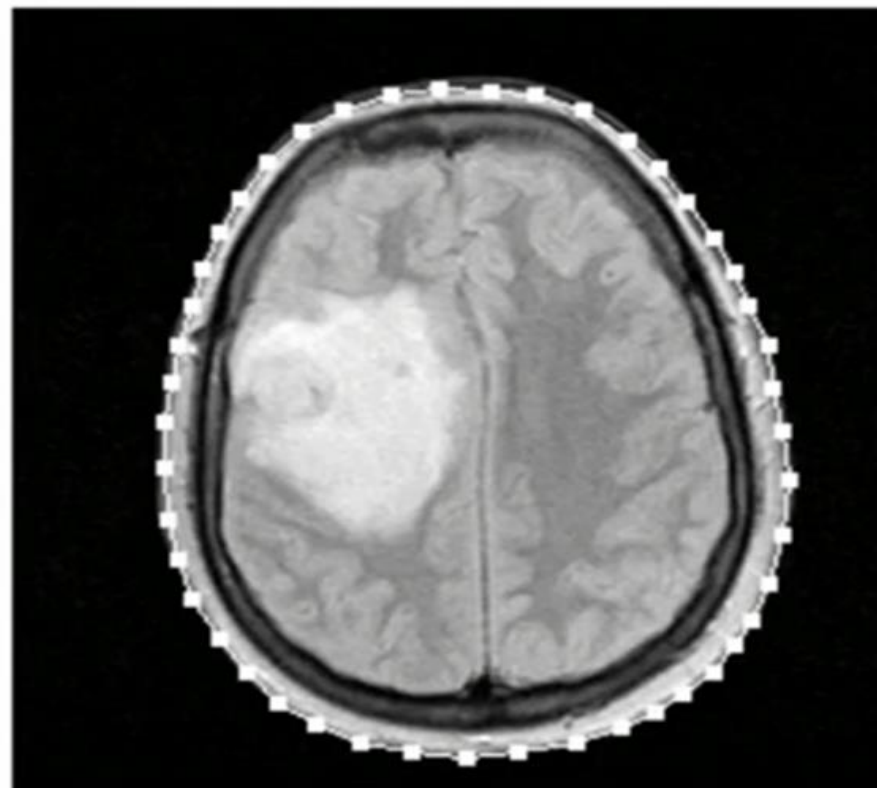
Active Contour (Snake Model)

★ Example

(a) Initial Snake

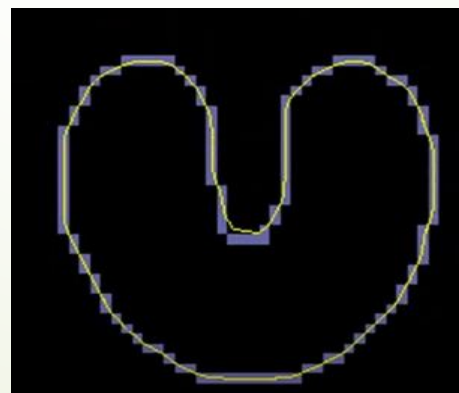
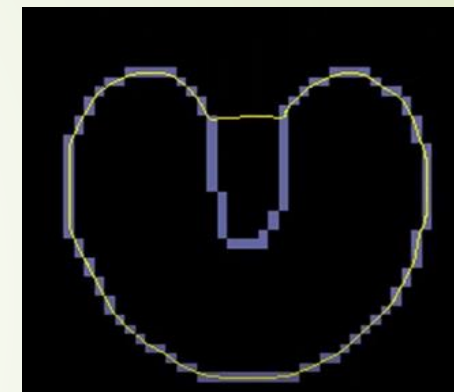
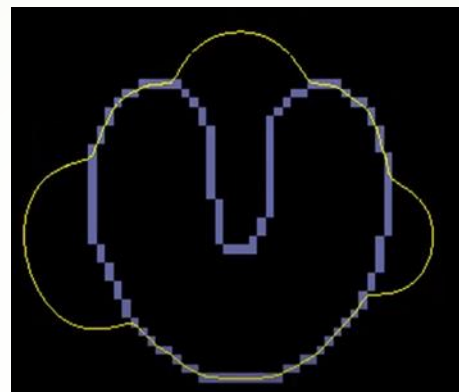


(b) Final Snake



★ Example

Active Contour (Snake Model)



Active Contour (Intelligent Scissors)

Intelligent Scissor

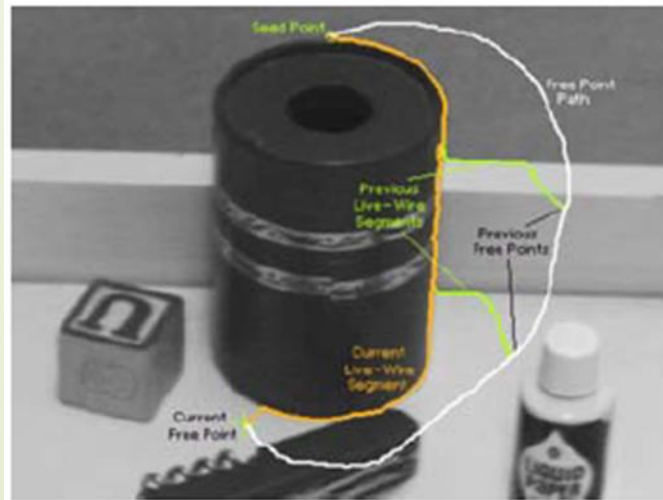
- ★ Intelligent Scissors algorithm (Intelligent Scissors) is an interactive image segmentation algorithm.
- ★ It is proposed in 1995 by the Eric N. Mortensen and William A. Barrett.
- ★ It can be used for 2D image segmentation
- ★ The algorithm can be used to assist a user to precisely outline the region of interest.

Active Contour (Intelligent Scissors)

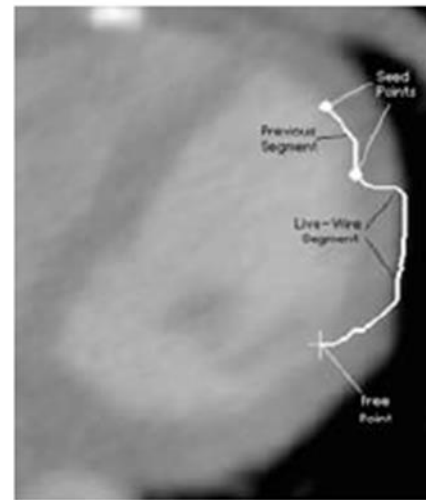
16

Intelligent Scissor

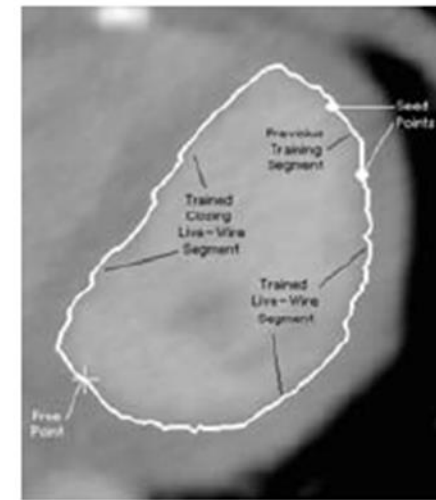
- ★ Active contours allow a user to roughly specify the boundary of interest.
- ★ The results may be unpredictable and may require additional user-based hints to achieve the desired results.
- ★ An alternative approach is to have the system optimize the contour in real time as the user is drawing.
- ★ As the user draws a rough outline (the white curve in), the system computes and draws a better curve that clings to high-contrast edges (the orange curve).



(a)



(b)

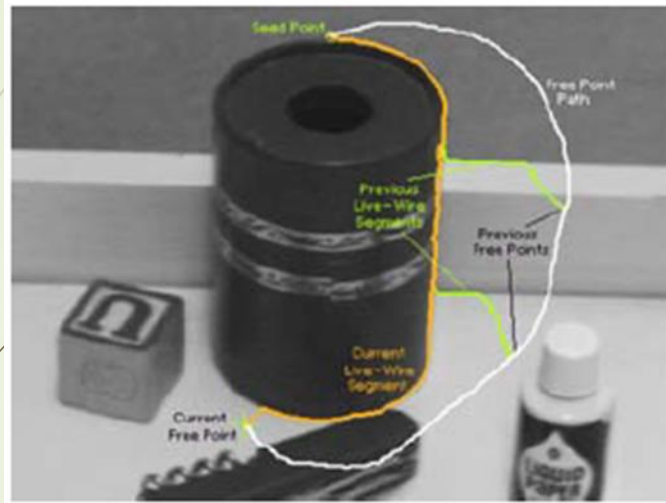


(c)

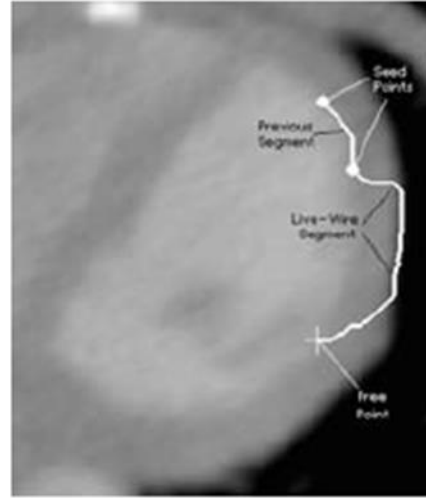
Active Contour (Intelligent Scissors)

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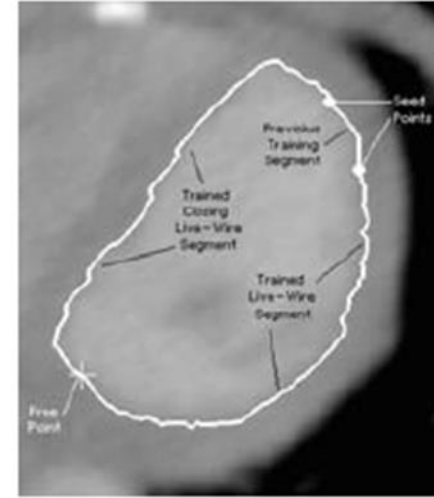
Intelligent Scissor



(a)



(b)



(c)

- (a) as the mouse traces the white path, the scissors follow the orange path along the object boundary (the green curves show intermediate positions)
- (b) regular scissors can sometimes jump to a stronger (incorrect) boundary;
- (c) after training to the previous segment, similar edge profiles are preferred

Active Contour (Intelligent Scissors)

To compute the optimal curve path (live-wire)

- ★ The image is first pre-processed to associate low costs with edges
 - (links between neighboring horizontal, vertical, and diagonal, i.e., N8 neighbors)
 - that are likely to be boundary elements.
- ★ This system uses a combination of
 - zerocrossing, gradient magnitudes, and gradient orientations to compute these costs.
- ★ Next, as the user traces a rough curve,
 - the system continuously re-computes the lowest cost path between the starting seed point and the current mouse location
 - using Dijkstra's algorithm

Active Contour (Intelligent Scissors)

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To compute the optimal curve path (live-wire)

- ★ A breadth-first dynamic programming algorithm that terminates at the current target location.
- ★ In order to keep the system from jumping around unpredictably,
 - the system will “freeze” the curve to date (reset the seed point) after a period of inactivity.
- ★ To prevent the live wire from jumping onto adjacent higher-contrast contours,
 - the system also “learns” the intensity profile under the current optimized curve, and
 - uses this to preferentially keep the wire moving along the same (or a similar looking) boundary.

Active Contour (Intelligent Scissors)

Extension

- ★ *Mortensen and Barrett (1999) use tobogganing,*
 - which is a simple form of watershed region segmentation
- ★ The resulting region boundaries are turned into a much smaller graph,
- ★ The Dijkstra algorithm is then run on this reduced graph, resulting in much faster (and often more stable) performance.
- ★ *Another extension to intelligent scissors is to use a probabilistic framework*
 - JetStream (Perez, Blake, and Gangnet 2001 ').
- ★ Instead of re-computing an optimal curve at each time instant,
 - a simpler system can be developed by simply “snapping” the current mouse position to the nearest likely boundary point
 - (Gleicher 1995).

Active Contour (Level Sets)

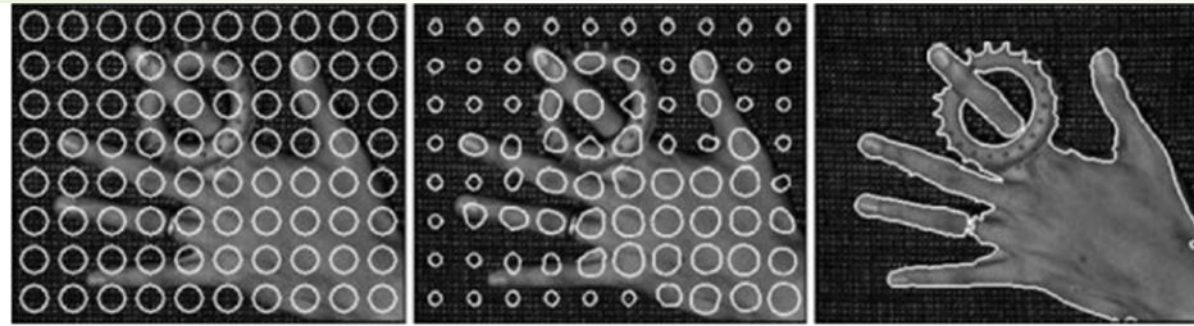
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Level Sets

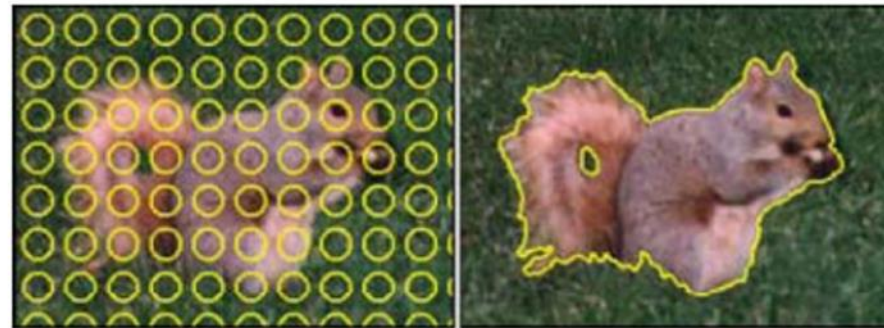
- ★ A limitation of active contours based on parametric curves of the form $f(s)$, e.g., snakes is that it is challenging to change the topology of the curve as it evolves.
- ★ If the shape changes dramatically, curve reparameterization may also be required.
- ★ An alternative representation for such closed contours is to use a level set,
 - Level Set evolve to fit and track objects of interest by modifying the underlying embedding function instead of curve function $f(s)$.

Active Contour (Level Sets)

Level Sets



(a)



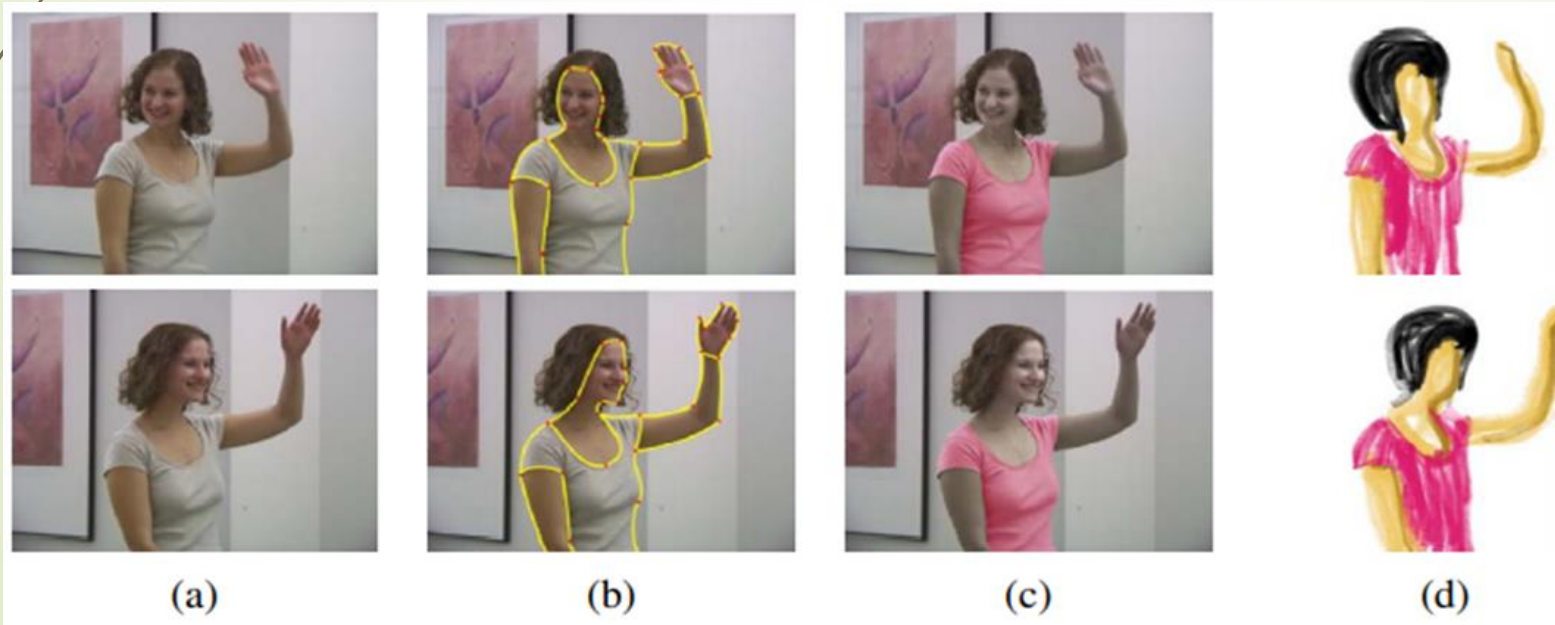
(b)

(a) grayscale image segmentation and (b) color image segmentation

Active Contour (Level Sets)

Level Sets: Application-Rotoscoping

- ★ Rotoscoping is an animation technique
 - in which animators trace over footage, frame by frame, for use in live-action and animated movies.
- ★ Originally, recorded live-action movie images were projected onto a frosted glass panel and re-drawn by an animator.
- ★ This projection equipment is referred to as a rotoscope.



(a) original frames;
(b) rotoscoped contours;
(c) re-colored blouse;
(d) rotoscoped hand-drawn animation

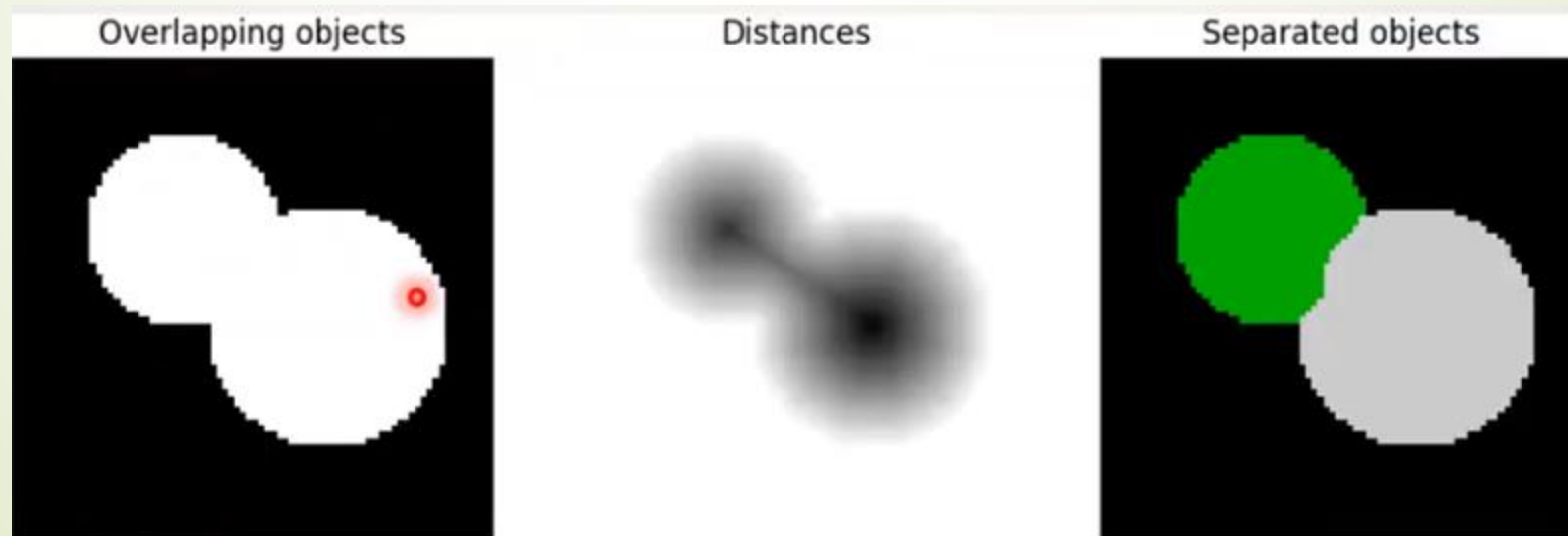
Active Contour (Level Sets)

Level Sets: Other Applications

- ★ They can be used to track facial features for performance-driven animation.
- ★ They can also be used to track heads and people, as well as moving vehicles.
- ★ Additional applications include medical image segmentation,
 - computerized tomography (3D medical imagery) or ultrasound scans.

Watershed Segmentation

- ★ The watershed algorithm is a classic algorithm used for segmentation and is especially useful when extracting *touching* or *overlapping objects* in images.
- ★ The watershed algorithm is based on extracting sure background and foreground and then using markers will make a watershed run and detect the exact boundaries.
- ★ For markers, it can be user-defined like
 - Manually clicking and getting the coordinates for markers
 - Using some defined algorithms such as *thresholding* or any *morphological operations*.

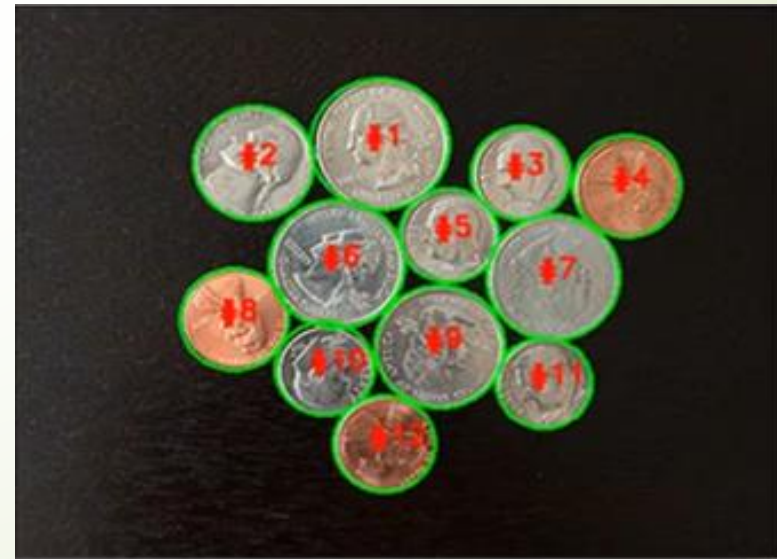


Watershed Segmentation

- ★ The watershed algorithm is used the following steps for image segmentation.
1. Finding the *sure background* using *morphological operations* like opening and dilation.
 2. Finding the *sure foreground* using *distance transform*.
 3. *Unknown area* (the area that neither lies in the foreground nor background) is used as *a marker* for the watershed algorithm.



Original image



Segmented Output image

Regions Based Segmentation

- ★ A region can be classified as a group of connected pixels exhibiting similar properties.
- ★ The similarity between pixels can be in terms of intensity, color, etc.
- ★ Region-based segmentation methods are preferred over edge-based segmentation methods in case of a *noisy image*.
- ★ Region-Based techniques are further classified into two types based on the approaches they follow.
 - Region growing method
 - Region *splitting and merging* method

Regions Based Segmentation

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Overview of Region Growing Method

- ★ Start with some pixel as the seed pixel and then check the adjacent pixels.
- ★ If the adjacent pixels abide by the predefined rules, then that pixel is added to the region of the seed pixel and
- ★ The process continues till there is no similarity left.
- ★ This method follows the *bottom-up approach*.
- ★ In case of a region growing, *the preferred rule can be set as a threshold*.
- ★ **Example:** Consider a seed pixel of 2 in the given image and a threshold value of 3, if $\text{pixel} > 3 \Rightarrow R2$, else $R1$.

1	1	5	6	5	5
2	1	6	7	4	6
3	2	7	4	6	7
1	0	5	5	7	6
2	0	4	6	8	5
0	1	6	4	5	8

Original Image

1	1	5	6	5	5
2	1	6	7	4	6
3	2	7	4	6	7
1	0	5	5	7	6
2	0	4	6	8	5
0	1	6	4	5	8

Region growing process with 2 as the seed pixel.

R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2
R1	R1	R2	R2	R2	R2

Splitting image into two regions based on a threshold.

Regions Based Segmentation

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Overview of Region Splitting

- ★ Splitting the image into successively finer regions is one of the oldest techniques in computer vision.
- ★ The basic idea of region splitting is to *break the image into a set of disjoint regions* which are coherent within themselves:
- ★ This process is repeated until regions are either fairly uniform or below a certain size.

Overview of Region Merging

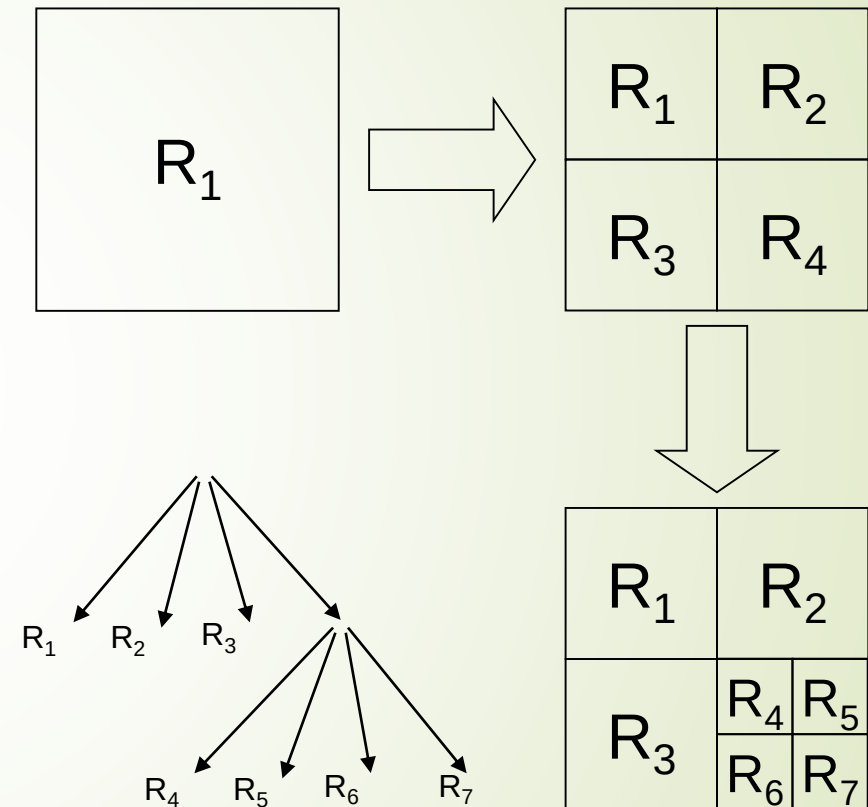
- ★ If only a splitting schedule is used then the final segmentation would probably contain many neighbouring regions that have identical or similar properties.
- ★ Thus, a merging process is used after each split which *compares adjacent regions and merges* them if necessary.
- ★ A region is selected as the seed region to check if adjacent regions are similar based on predefined rules.
- ★ If similar, they are merged into a single region.

Regions Based Segmentation

Overview of Region Splitting

★ Example:

- ★ One initial set that includes the whole image.
- ★ Similarity criteria.
- ★ Iteratively split regions into sub-regions.
- ★ Stop when no more splitting are possible.

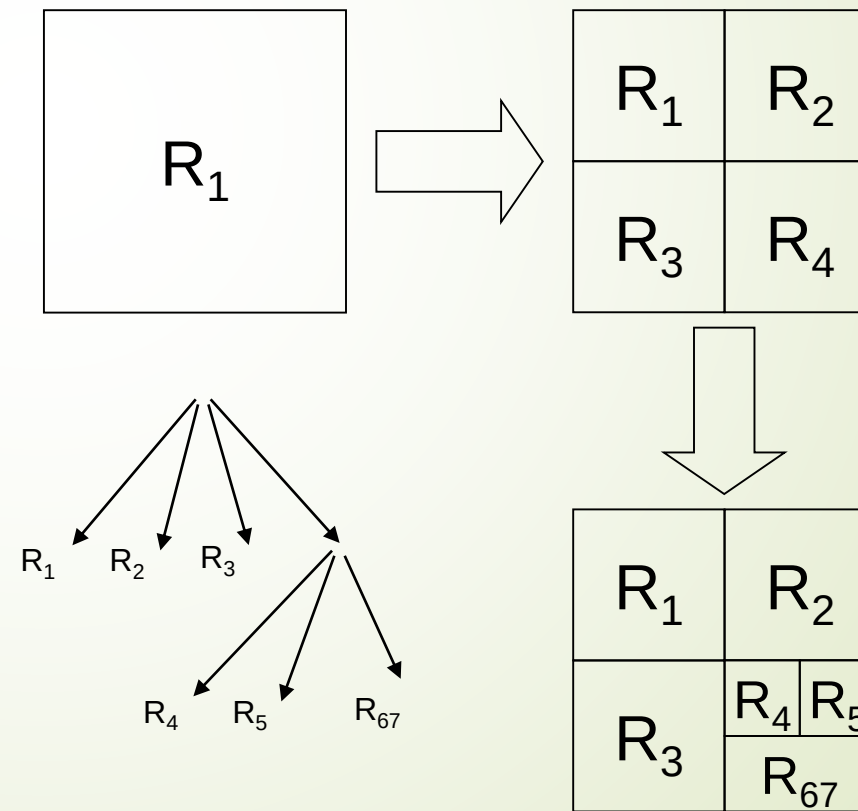


Regions Based Segmentation

Overview of Region Merging

★ Example:

- ★ Now assume that all pixels within each region are similar with respect to that region.
- ★ After comparing the split regions, regions R6 and R7 are found to be identical.
- ★ These are thus merged together.



Regions Based Segmentation

Region Splitting and Merging

- ★ Both region splitting and region merging are iterative processes.
- ★ Usually, first region splitting is done in image.
- ★ The image is successively *split into quadrants based on a homogeneity criterion*.
- ★ And then similar regions are *merged in order to form a good segmented image* of the original image.
- ★ Algorithms of this nature are called **split and merge algorithm**.
 1. *Define the criterion to be used for homogeneity*
 2. *Split the image into equal size regions*
 3. *Calculate homogeneity for each region*
 4. *If the region is homogeneous, then merge it with neighbors*
 5. *The process is repeated until all regions pass the homogeneity test*

Regions Based Segmentation

Split and Merge

★ Splitting techniques disadvantage:

- They create regions that may be adjacent and homogeneous, but not merged.

★ Split and Merge method:

- It is an iterative algorithm that includes both splitting and merging at each iteration.
- If a region R is inhomogeneous ($P(R) = \text{False}$) then is split into four sub regions
- If two adjacent regions R_i, R_j are homogeneous ($P(R_i \cup R_j) = \text{TRUE}$), they are merged
- The algorithm stops when no further splitting or merging is possible.

Regions Based Segmentation

Split and Merge: Application

- ★ The algorithm produces more compact regions than the pure splitting algorithm.
- ★ 3D – Imaging : A basic task in 3-D image processing is the segmentation of an image which classifies voxels/pixels into objects or groups. .
- ★ Several applications in the field of Medicine like magnetic resonance imaging (MRI).

Regions Based Segmentation

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Split and Merge



Result_ Region Split

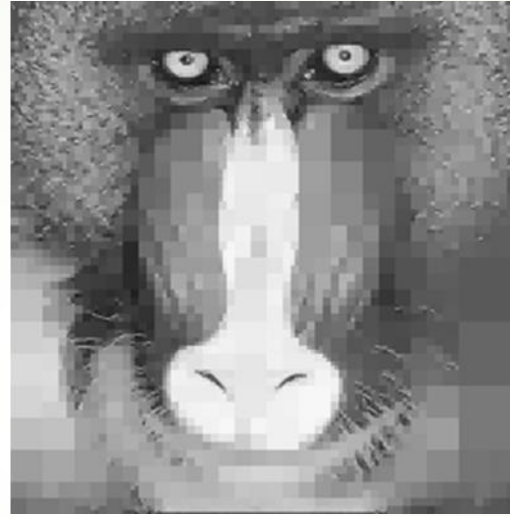
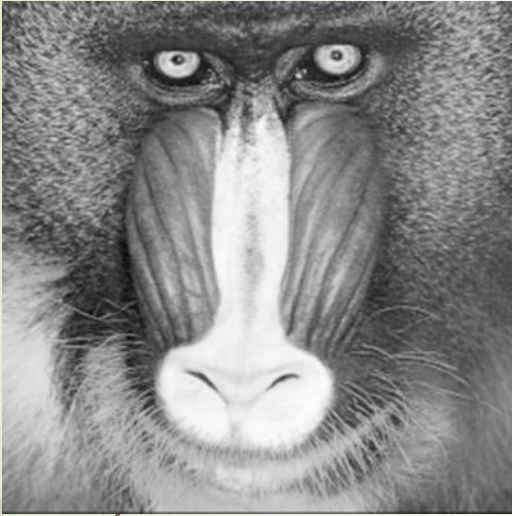


Result_ Region Split and Merge

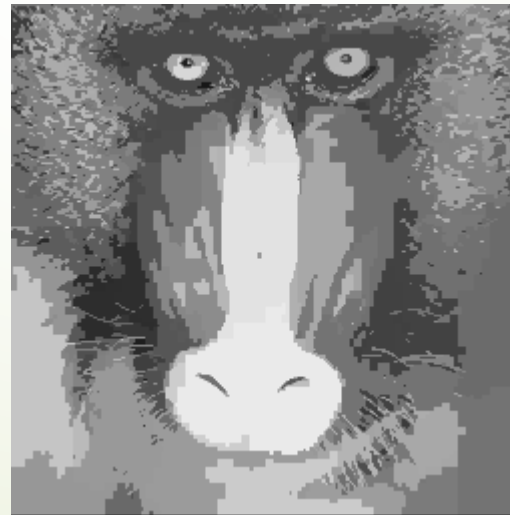
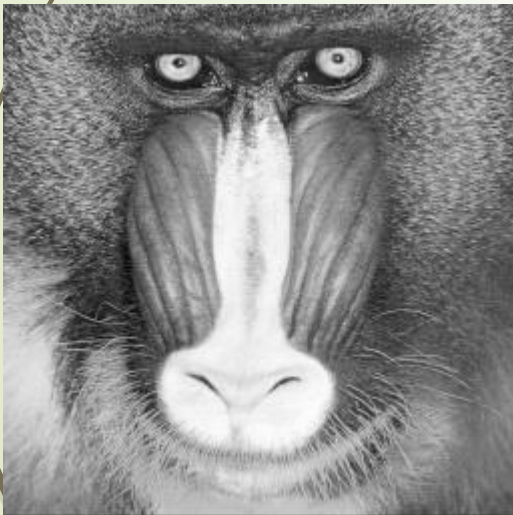
Regions Based Segmentation

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Split and Merge



Result_ Region Split



Result_ Region Split and Merge

Regions Based Segmentation

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Question in Region Splitting

- ★ Apply the region splitting and merging on the following image. Assume the threshold value be ≤ 3 .

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
0	2	1	0	2	3	5	4

Regions Based Segmentation

Question in Region Splitting

- ★ Apply the region splitting and merging on the following image. Assume the threshold value be ≤ 3 .

Solution:

- ★ If the absolute value of the difference of the maximum and minimum pixel intensities in a region is less than or equal to a threshold value, then the region does not require further splitting.

$$|Z_{\max} - Z_{\min}| \leq \text{threshold}$$

$Z_{\max}$ = Maximum pixel intensity value in a region.

$Z_{\min}$ = Minimum pixel intensity value in a region

Regions Based Segmentation

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Solution : Splitting

a									b
	5	6	6	6	7	7	6	6	
	6	7	6	7	5	5	4	7	
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	4	7	
	0	0	0	0	2	2	5	6	
	1	1	0	1	0	3	4	4	
	0	2	1	0	2	3	5	4	
				c					d

$$7 - 0 = 7$$

$7 - 4 = 3$				$7 - 2 = 5$					
R1	5	6	6	6	7	7	6	6	R2
	6	7	6	7	5	5	4	7	
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
R3	0	3	2	3	3	2	4	7	R4
	0	0	0	0	2	2	5	6	
	1	1	0	1	0	3	4	4	
	0	2	1	0	2	3	5	4	
$3 - 1 = 2$				$7 - 0 = 7$					

Regions Based Segmentation

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Solution: Splitting

$7 - 4 = 3$

a

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
0	2	1	0	2	3	5	4

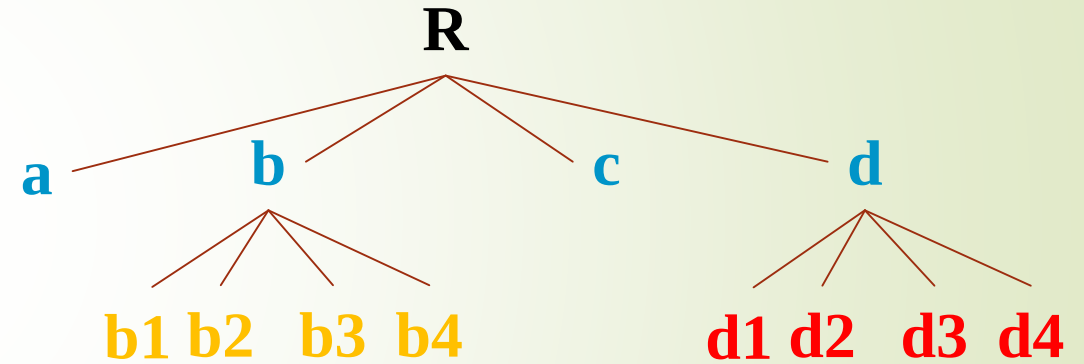
c

$3 - 1 = 2$

b1 **b2** **b3** **b4** **d1** **d2** **d3** **d4**

b

d



$$b1 \Rightarrow 7 - 5 = 2$$

$$b2 \Rightarrow 7 - 4 = 3$$

$$b3 \Rightarrow 3 - 2 = 1$$

$$b4 \Rightarrow 6 - 4 = 2$$

$$d1 \Rightarrow 3 - 2 = 1$$

$$d2 \Rightarrow 7 - 4 = 3$$

$$d3 \Rightarrow 3 - 0 = 3$$

$$d4 \Rightarrow 5 - 4 = 1$$

Regions Based Segmentation

Solution: Merging

★ Check the adjacent regions, if they are falling within the threshold, then merge.

Consider region **a** and **b1**

Max = 7, Min = 4

$|7 - 4| = 3 \leq 3$ (**Merge**)

Consider region **a-b1** and **b2**

Max = 7, Min = 4

$|7 - 4| = 3 \leq 3$ (**Merge**)

Consider region **a-b1-b2** and **b4**

Max = 7, Min = 4

$|7 - 4| = 3 \leq 3$ (**Merge**)

Consider region **a-b1-b2-b4** and **d2**

Max = 7, Min = 4

$|7 - 4| = 3 \leq 3$ (**Merge**)

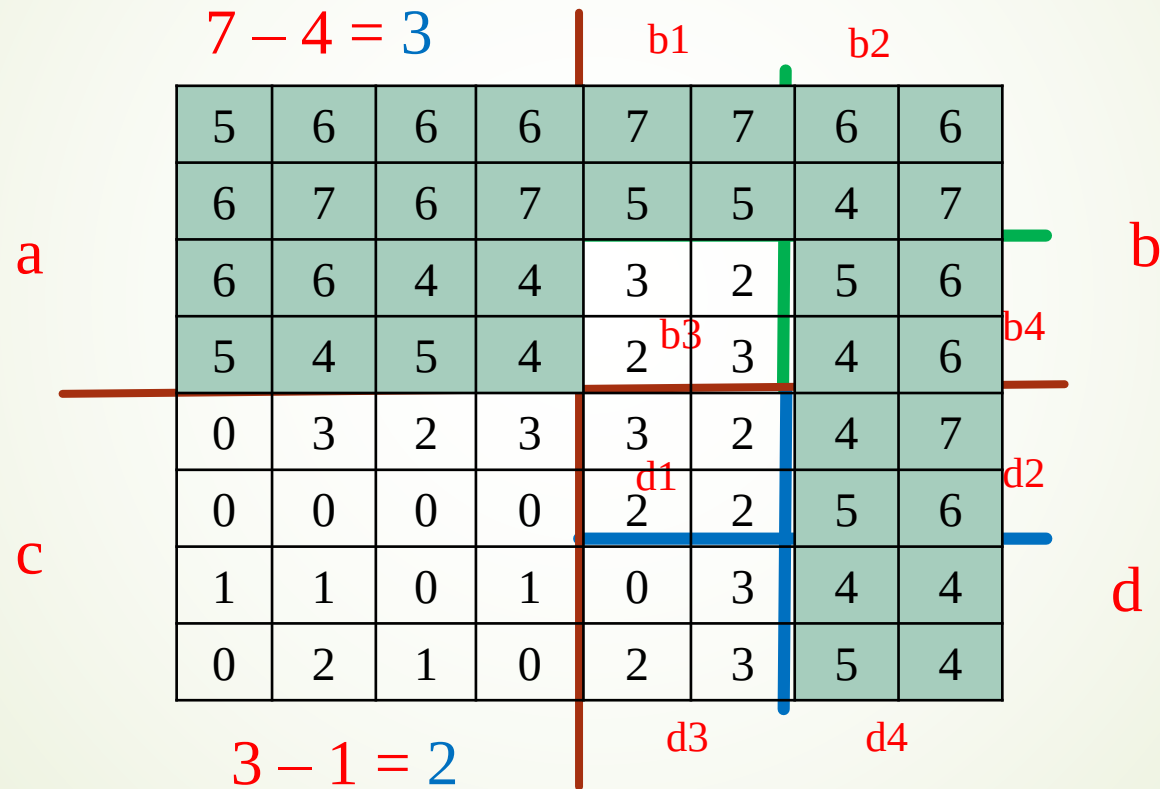
Consider region **a-b1-b2-b4-d2** and **d4**

Max = 7, Min = 4

$|7 - 4| = 3 \leq 3$ (**Merge**)

Regions Based Segmentation

Solution: Merging



Regions Based Segmentation

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Question

Apply the region growing on the following image with initial point at **(2, 2)** and threshold value as **2**. Use 4 connectivity.

$$\text{Abs}(\text{Seed} - \text{pixel}) \leq 2$$

$$7 - 7 \leq 2$$

$$7 - 6 \leq 2$$

$$7 - 5 \leq 2$$

0	1	2	0
2	5	6	1
1	4	7	3
0	2	5	1

0	1	2	0
2	5	6	1
1	4	7	3
0	2	5	1

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Regions Based Segmentation

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Question

Apply the region growing on the following image with seed point as 6 and threshold value as 3. Use 8 way connectivity.

$$\text{Abs}(\text{Seed} - \text{pixel}) \leq 3$$

$$6 - 7 < 3$$

$$6 - 6 < 3$$

$$6 - 5 < 3$$

$$6 - 4 < 3$$

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	1	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1

Regions Based Segmentation

Kernel Density Estimation

- ★ Mean shift is based on the concept of Kernel Density Estimation(KDE), which is a way to estimate the probability density function of a random variable.
- ★ It works by providing weights to each data point.
- ★ The weight function is called a *kernel*.
- ★ There are many kinds of kernels, one is the Gaussian kernel.
- ★ Adding all those kernels together creates a density function(probability surface).
- ★ The resultant density function variation depends on the used *bandwidth parameter*.

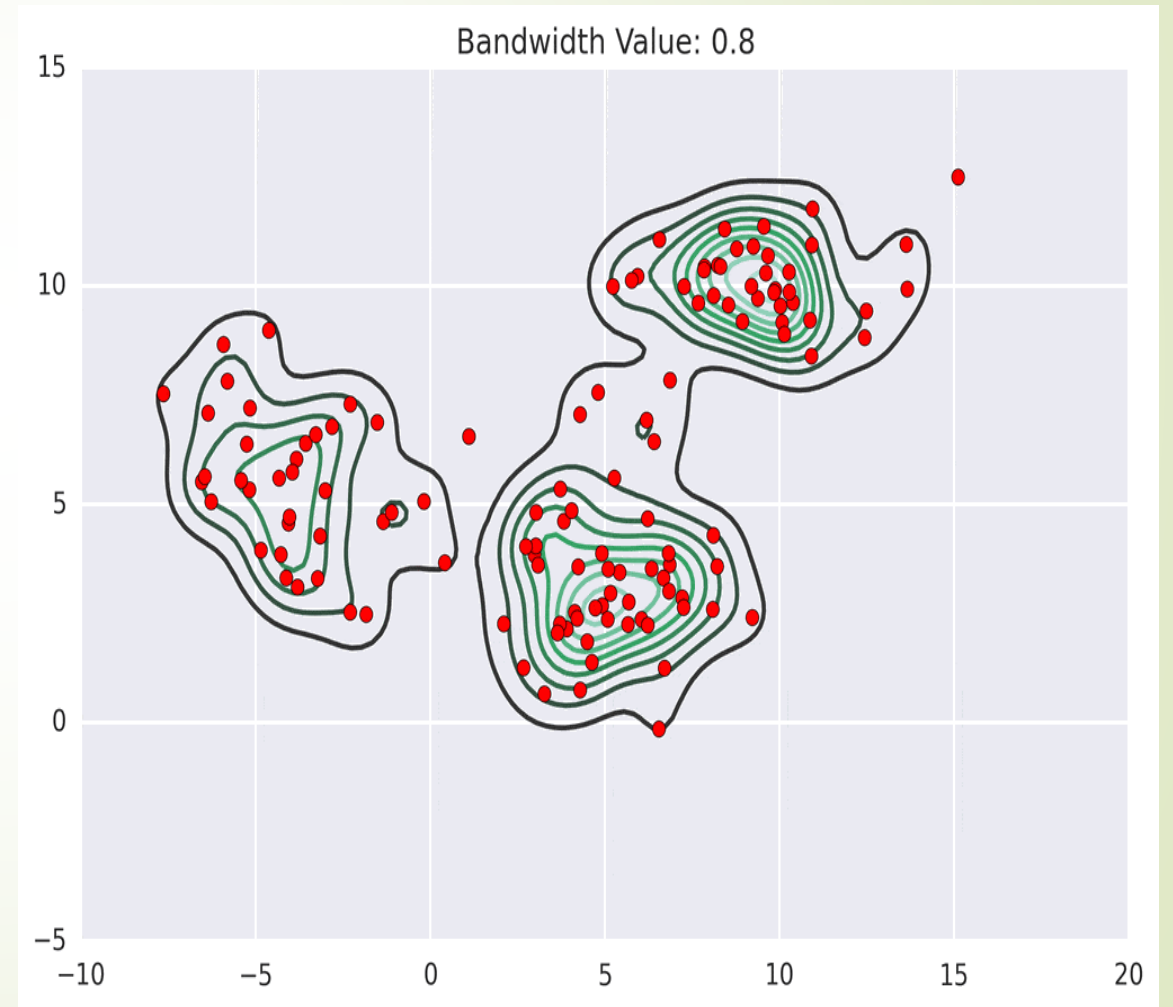
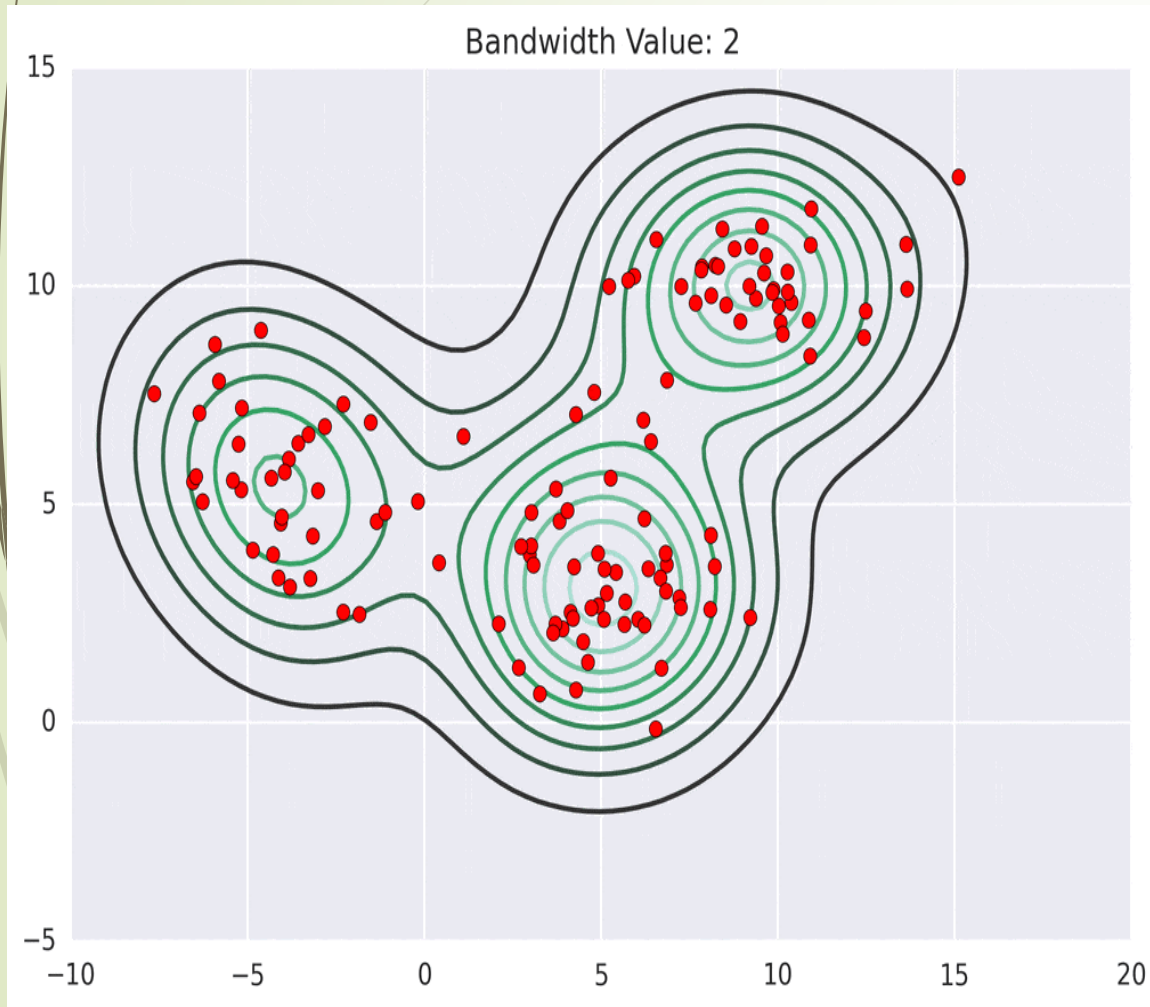
Regions Based Segmentation

Mean Shift Segmentation

- ★ Mean shift is based on the idea of KDE, but what makes it different is that using the bandwidth parameter.
- ★ The points can be made climb uphill to the nearest peak on the KDE surface.
- ★ So, iteratively shifting each point to climb uphill to the peak.
- ★ The bandwidth parameter used to make the KDE surface varies on the different sizes.
- ★ A small kernel bandwidth makes the KDE surface hold the peak for every data point more formally, saying each point has its cluster; on the other hand, large kernel bandwidth results in fewer kernels or fewer clusters.

Regions Based Segmentation

Mean Shift Segmentation



Regions Based Segmentation

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Idea of Mean Shift

- ★ Each hill represents a cluster.
- ★ Peak (mode) of hill represents “center” of cluster.
- ★ Each pixel climbs the steepest hill within its neighborhood.
- ★ Pixel is assigned to the hill (cluster) it climbs.

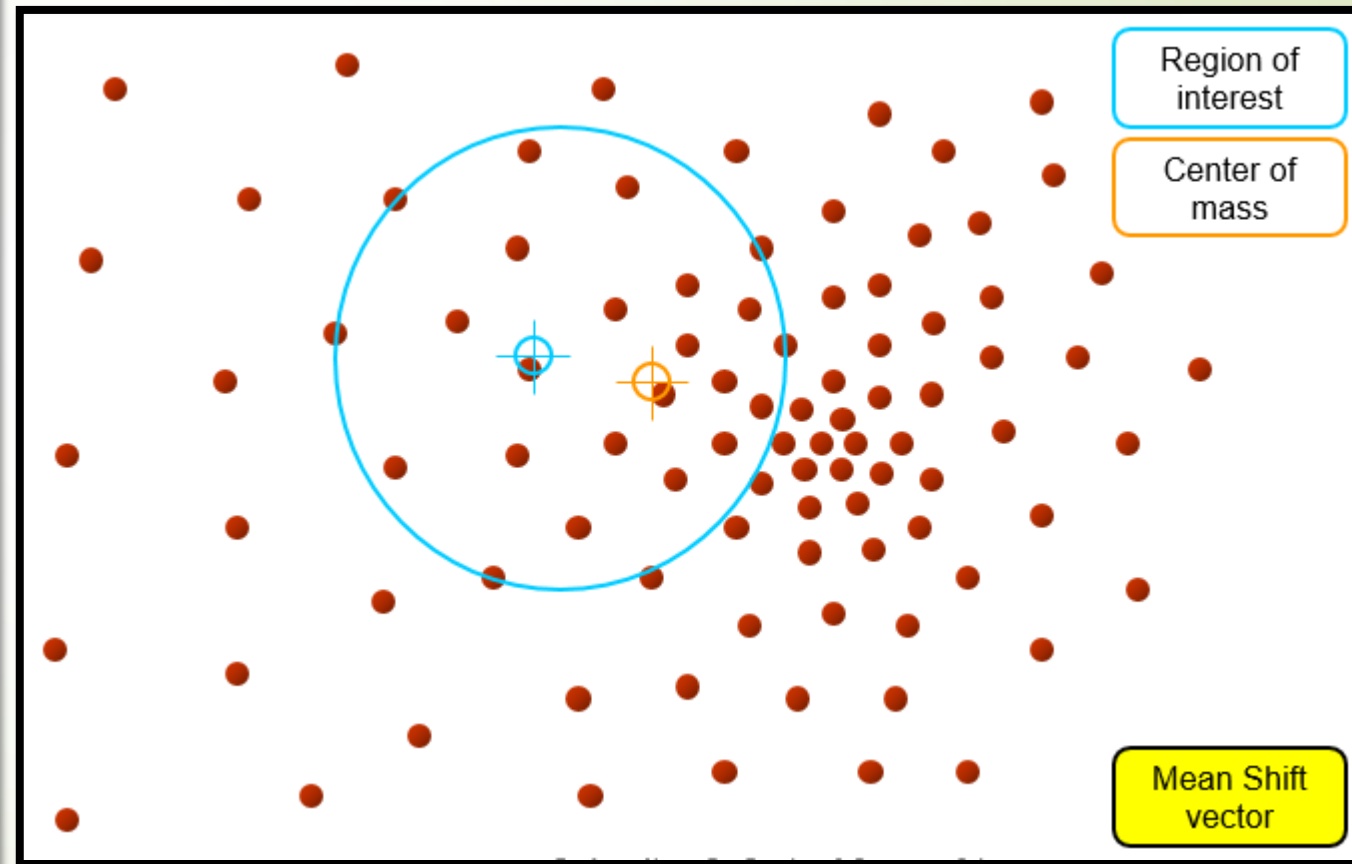
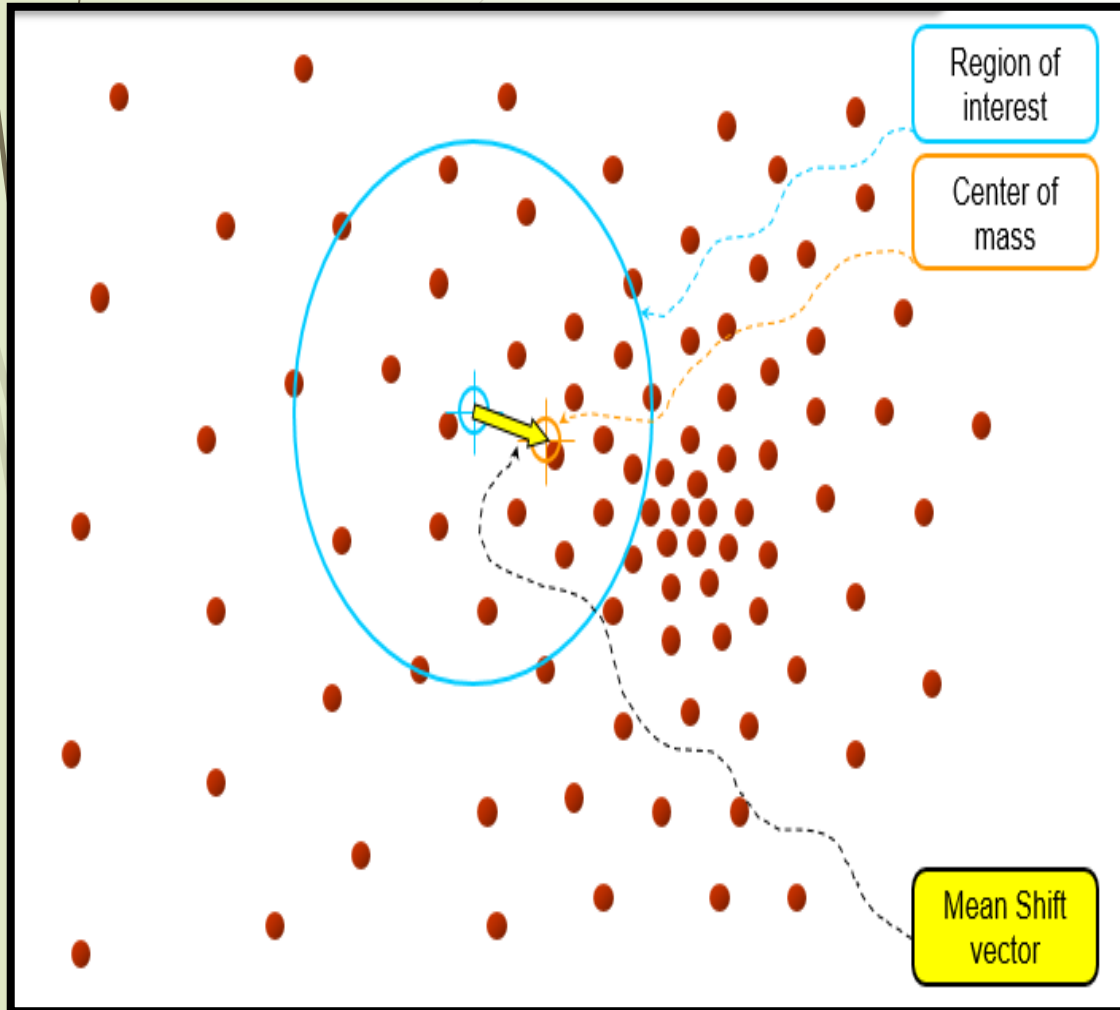
Hill Climbing using Mean Shift

- ★ Compute the centroid or mean within the Window size W .
- ★ Use the Weighted Means.
- ★ “Shift” the Window to the centroid.
- ★ Repeat ... Until Convergence.
- ★ Declare mode and assign it a cluster label.
- ★ Repeat for all pixel.
- ★ Feature that converge to same mode belong to same cluster

Regions Based Segmentation

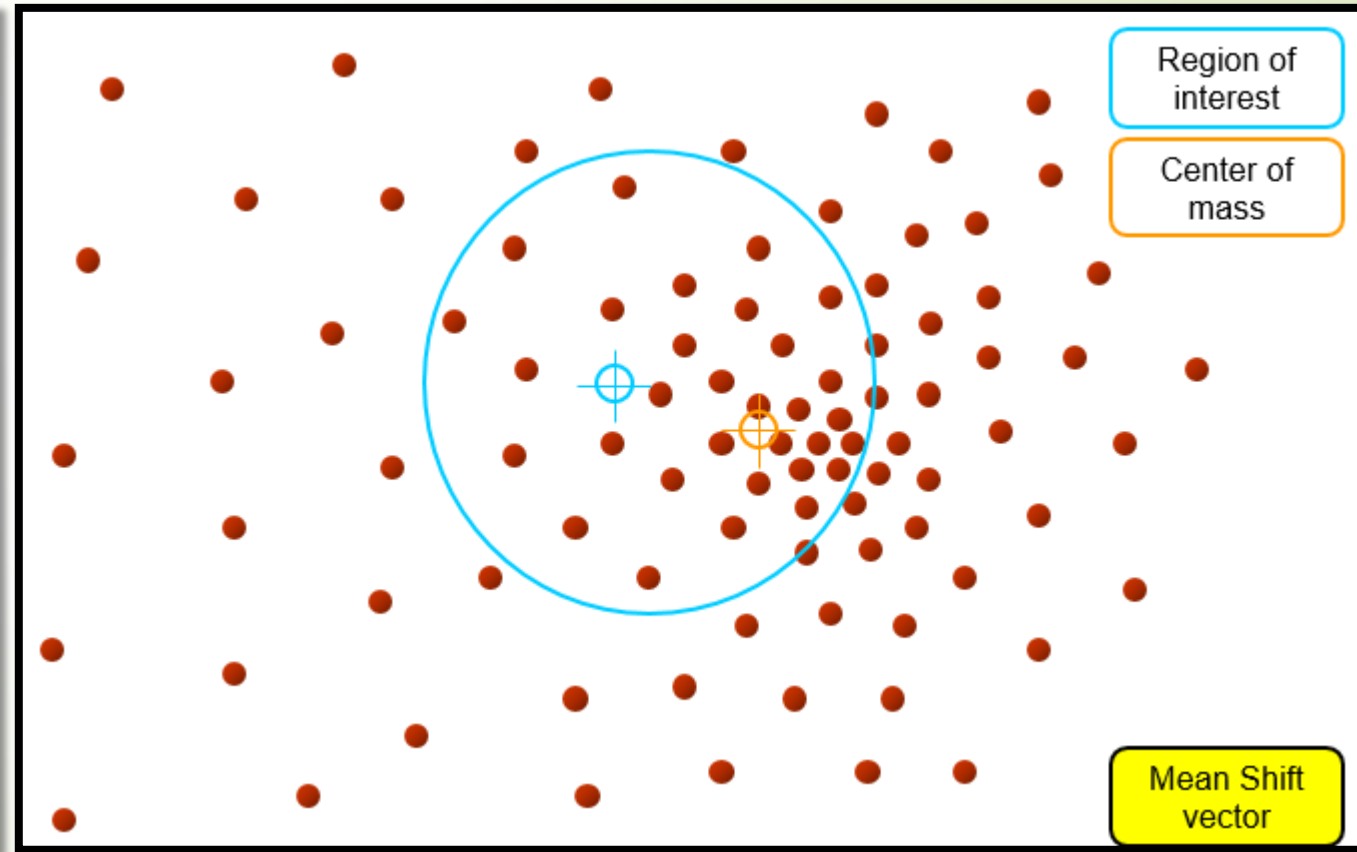
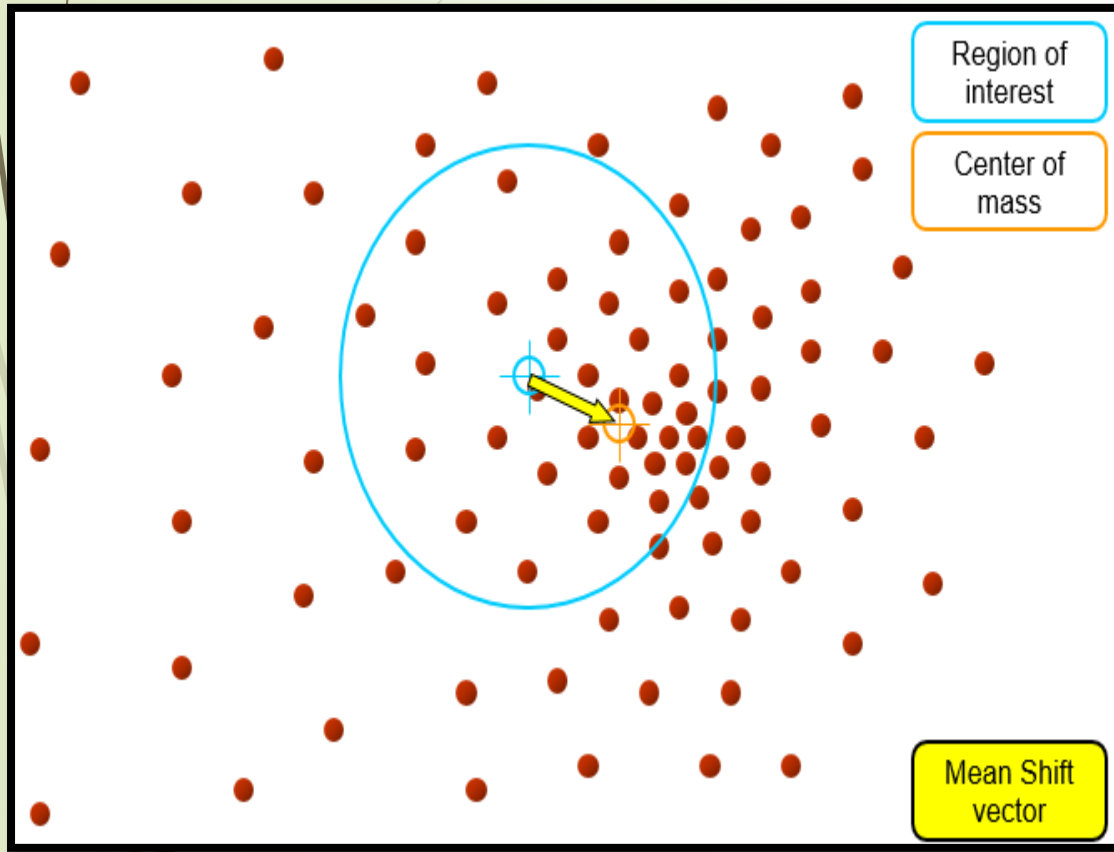
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Mean Shift Segmentation



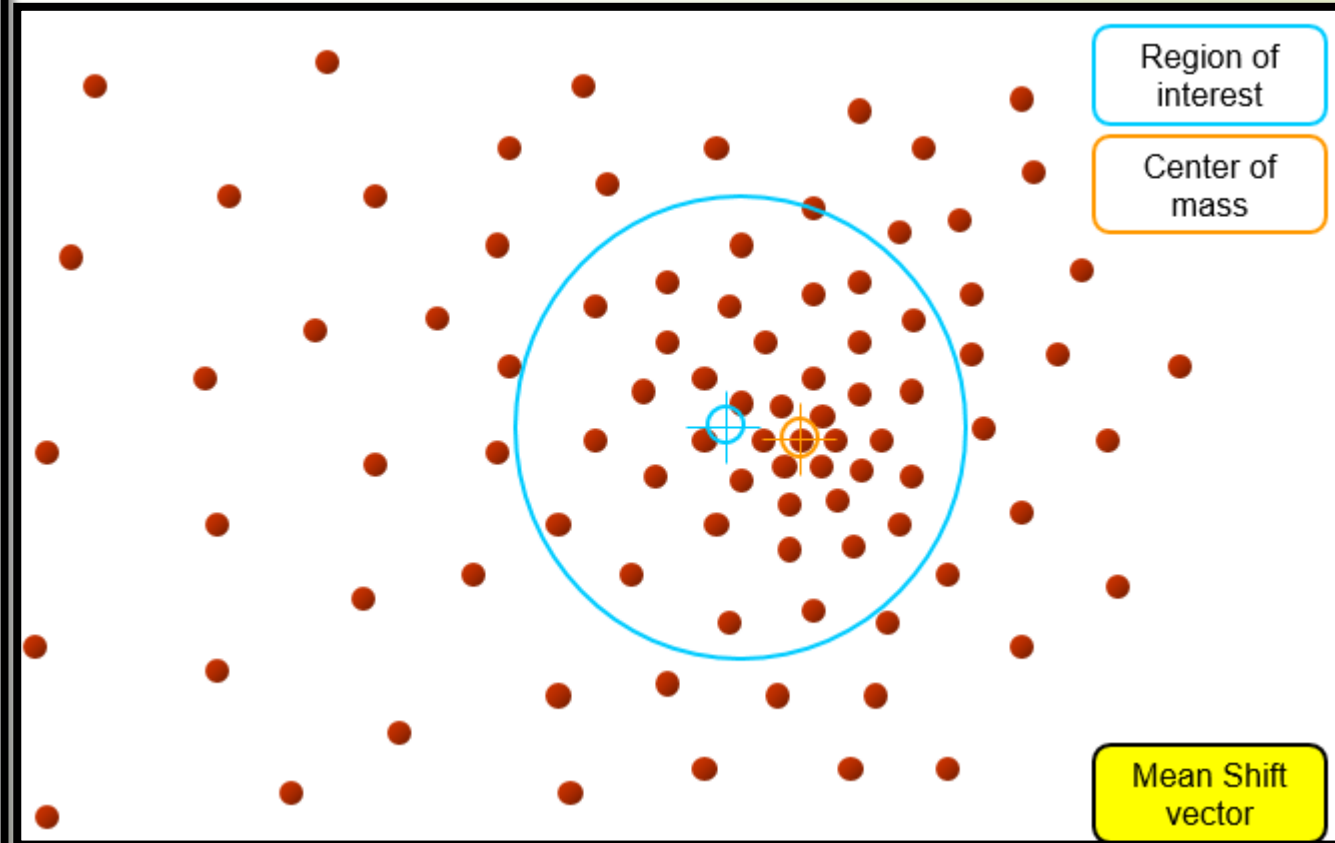
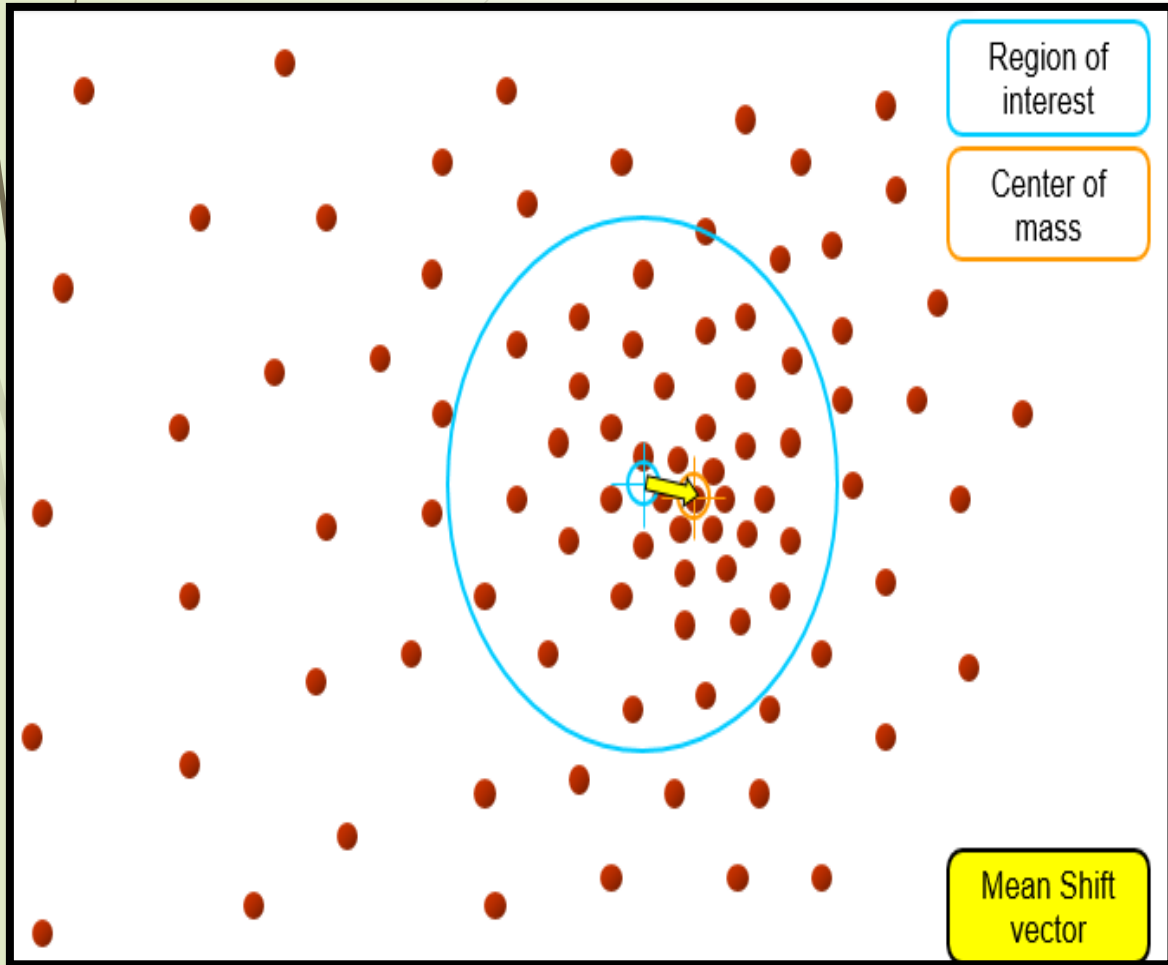
Regions Based Segmentation

Mean Shift Segmentation



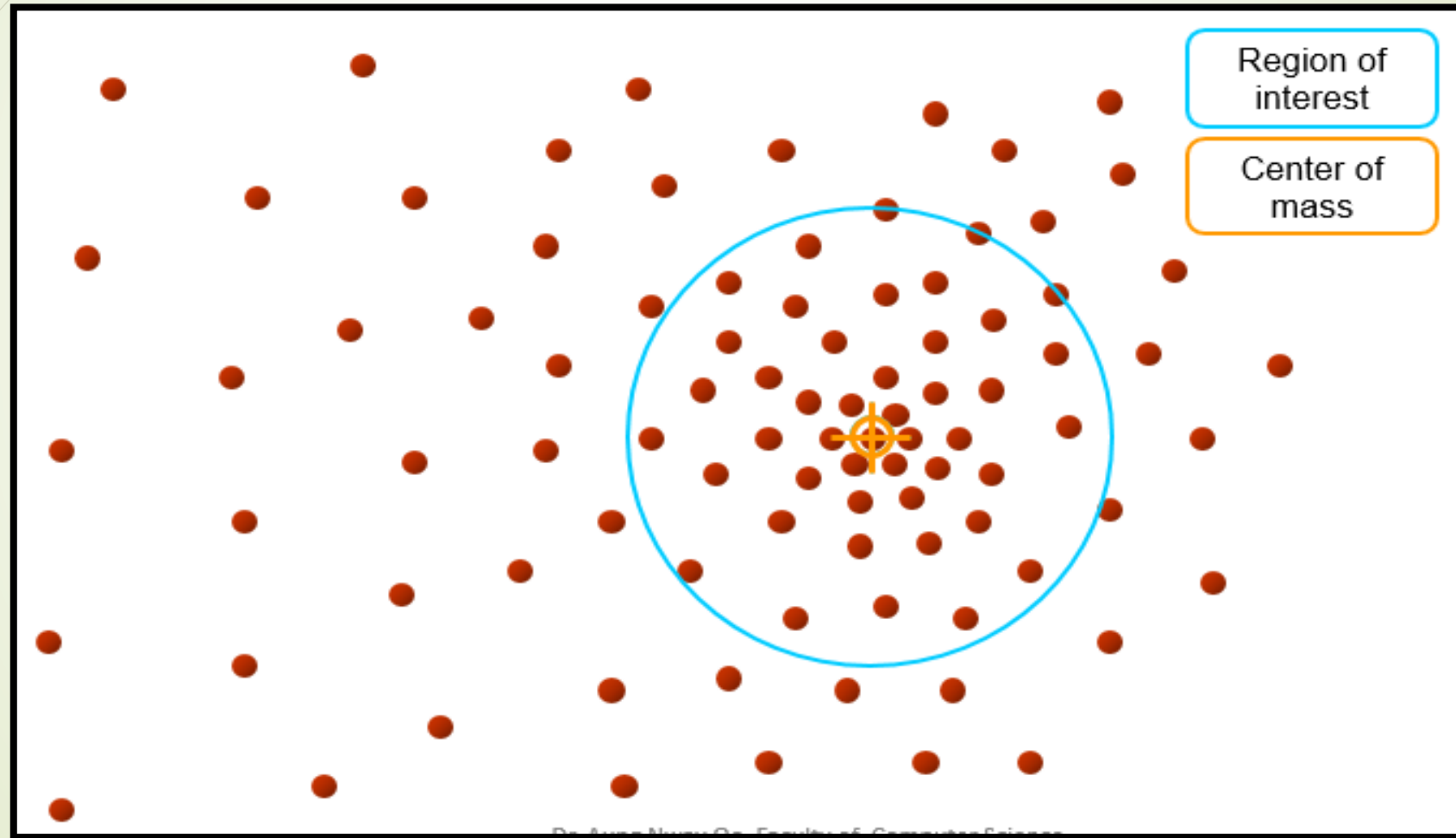
Regions Based Segmentation

Mean Shift Segmentation



Regions Based Segmentation

Mean Shift Segmentation



Regions Based Segmentation

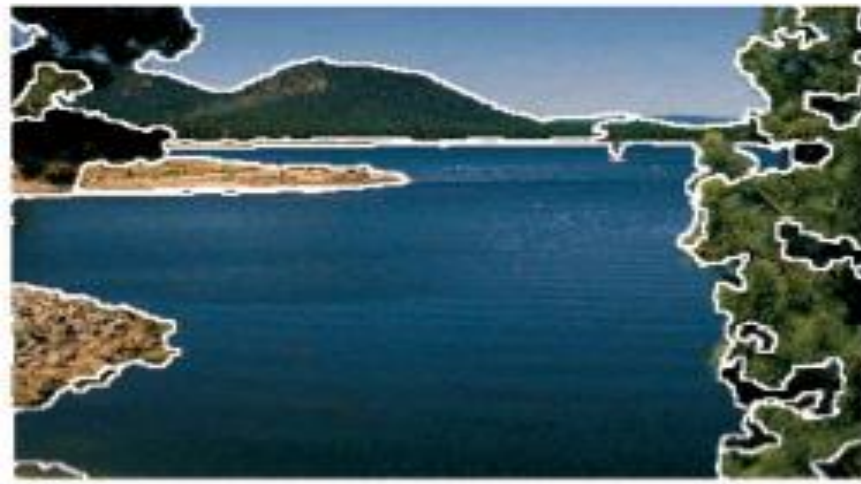
Mean Shift Algorithm

★ Clustering:

- 1: Set $m_i = f_i$ as initial mean for each pixel i .
- 2: Repeat the following for each mean m_i :
 - a: Place the window of size W around m_i .
 - b: Compute centroid m within the window. Set $m_i = m$.
 - c: Stop if shift in mean m_i is less than a threshold then m_i is the mode.
- 3: Label all pixels that have same mode as belonging to the same cluster.

Regions Based Segmentation

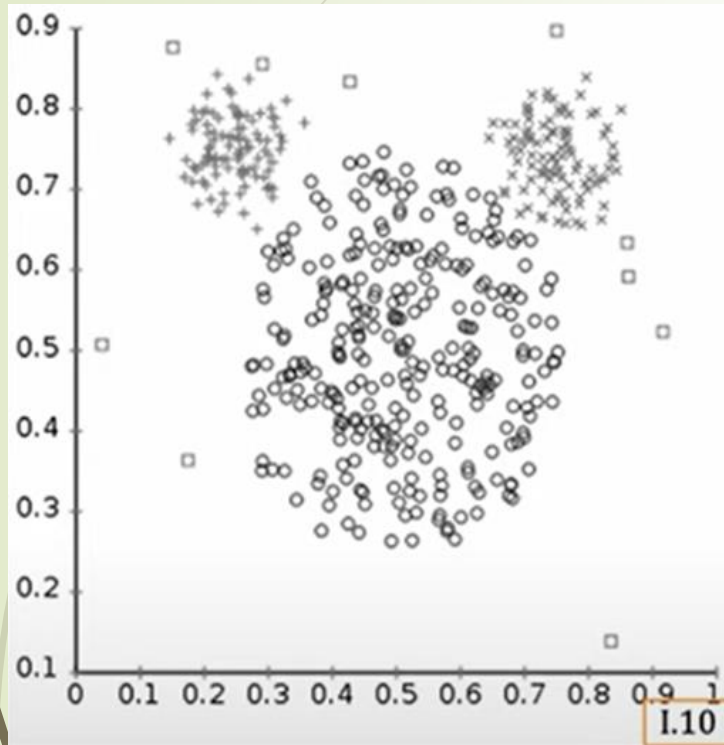
Mean Shift Algorithm: Example



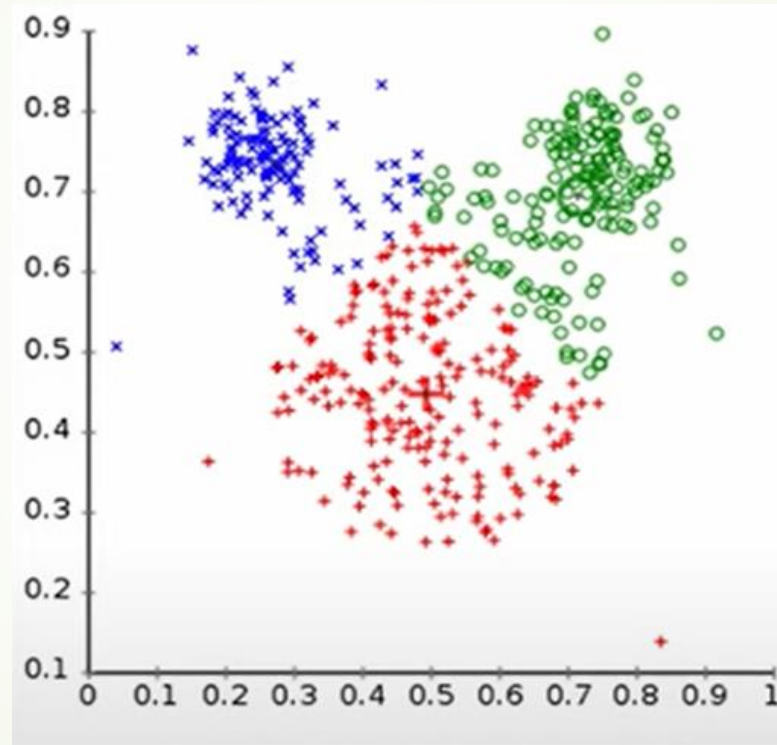
Regions Based Segmentation

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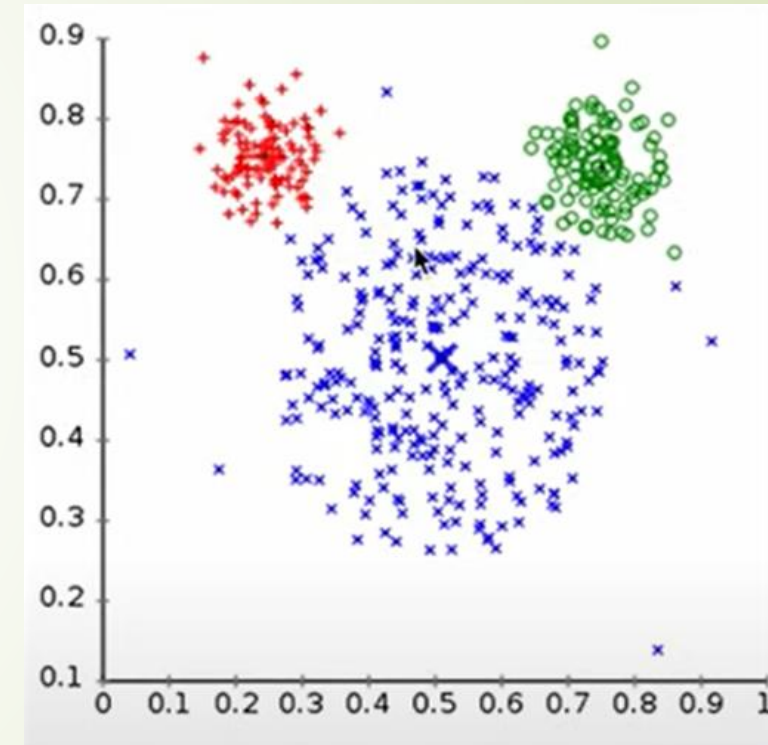
K-means Vs Mean Shift



Original Data



K-Means (k=3)



Mean Shift

Regions Based Segmentation

K-means Vs Mean Shift



Original Data



K-Means (k=3)



Mean Shift